Student Number:									

Surname:																
First Name(s):																

Department of Mathematics and Applied Mathematics

MAM2000W: 2LA (Linear Algebra)

Test 1 **10 March 2020**

Time : 1.5 hours **Maximum marks: 51**

Full marks: 50

- This question paper consists of 10 pages (including this one).
- Answer all questions in the spaces provided on this question paper. Use the backs of pages when you run out of space.
- Use your *UCT Examination Answer Book* for rough work. We won't take this work in! Only the answers that you write on this question paper will be marked.
- Calculators are not allowed. When your answer contains constants such as π, e or $\sqrt{2}$, leave them in that form. Don't simplify your answers.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.

DO NOT WRITE BELOW THIS LINE

Q1	Q2	Q3	Q4	Q5	Q6	Q7	TOTAL
10	6	11	7	8	4	5	**51**

Student Number:									

Question 1. [10 points] Let $A = \begin{pmatrix} 0 & 8 & 2 \\ 4 & 14 & 0 \\ -8 & -12 & 4 \end{pmatrix}$.

(a) Find elementary matrices E_1, E_2 and E_3 and a matrix U in row echelon form such that $E_3E_2E_1A = U$.

Student Number:									

(b) Find the the set of solutions of the equation $A\boldsymbol{x} = \mathbf{0}$.

(c) Answer the following questions, giving reasons for your answers:

(i) Can A be written as a product of elementary matrices?

(ii) Is the matrix U non-singular?

Student Number:									

Question 2. **[6 points]** In this question, you are given the augmented matrices of two linear systems, already in row echelon form. Write down the **solution set** of the system in each case.

(a)

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & 9 \\ 0 & 1 & 3 & 8 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

(b)

$$\left(\begin{array}{cccc|c} 1 & 3 & 5 & 7 & 2 \\ 0 & 0 & 1 & 1 & 4 \end{array}\right)$$

Student Number:									

Question 3. [11 points] Let

$$A_1 = \begin{pmatrix} 2 & 0 & -4 \\ 0 & -3 & 0 \\ 0 & 0 & -7 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 3 & 1 \end{pmatrix},$$

$$A_4 = \begin{pmatrix} 1 & -3 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 7 \end{pmatrix}, \quad A_5 = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad A_6 = \begin{pmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

In each of (a) to (g) below, a description of a matrix is given. In each case, choose **one** matrix from the list above satisfying the description given. Write **only one of** A_1, A_2, A_3, A_4, A_5, A_6 in the appropriate block on the answer sheet. If it is impossible to find a matrix satisfying the description given, write "Impossible". You may use a matrix more than once, or not at all.

(a) A matrix that is not row-equivalent to I_3.

(b A symmetric matrix in row echelon form.

(c) A singular elementary matrix.

(d) An upper triangular matrix which is invertible.

(e) An elementary matrix that is not in row echelon form.

(f) A matrix which can be written as the product of two elementary matrices, neither of which is an identity matrix.

Also give the two elementary matrices, in the correct order.

(g) A matrix which is not elementary and not in row echelon form, but can be written as the product of an elementary matrix and a matrix in row echelon form.

Give the two matrices in the product, in the correct order.

Student Number:									

Question 4. [**7 points**] Consider the statement:

For every $n \times n$ matrix A, if $A^2 = A$ and $A \neq I_n$, then A is not invertible. $(*)$

(a) Write down the converse of the statement $(*)$.

(b) Write down the contrapositive of the statement $(*)$.

(c) Prove or disprove the statement $(*)$.

(d) Prove or disprove the converse of the statement $(*)$.

Student Number:									

Question 5. [**8 points**] Read the following result and its proof and then answer the questions following it.

Theorem 1. *Every elementary matrix is invertible, and its inverse is also an elementary matrix.*

Proof. Let E be an $n \times n$ elementary matrix.
Then $E = e(I_n)$, for some elementary row operation e. (1)
Let e^{-1} be the inverse operation of e. (2)
Then $E \cdot e^{-1}(I_n) = e(I_n) \cdot e^{-1}(I_n) = e(e^{-1}(I_n)) = I_n$ (3)
and $e^{-1}(I_n) \cdot E = e^{-1}(E) = e^{-1}(e(I_n)) = I_n$. (4)
It follows that E is invertible, (5)
and that E^{-1} is also an elementary matrix. (6)

$\square$

(a) If e is an elementary row operation and A an $n \times n$ matrix, what does the symbol $e(A)$ stand for?

(b) There is a matrix C such that $e(A) = CA$. How is the matrix C obtained?

(c) Use your answer to (b) to explain why we can say in (3) that
$e(I_n) \cdot e^{-1}(I_n) = e(e^{-1}(I_n))$.

Student Number:									

(d) Why can we say at (5) that E is invertible?

(e) How does it follow at (6) that E^{-1} is elementary?

(f) Can an elementary matrix be its own inverse? Give a reason for your answer.

(g) Prove or disprove: The inverse of the product of two elementary matrices is an elementary matrix.

Student Number:									

Question 6. [**4 points**] For each of the following statements, say whether it is true or false. If it is true, prove it; if it is false, give a counter example and explain why it is one.

(a) The sum of two elementary matrices is always an elementary matrix.

(b) For any invertible matrices A and B, A^7B^5 is invertible.

Question 7. [**5 points**] For any $n \times n$ matrix $C = (c_{ij})$, the *trace* of C, denoted by $\mathrm{tr}(C)$, is defined by

$$\mathrm{tr}(C) = \sum_{i=1}^{n} c_{ii}.$$

(a) Prove that for any $n \times n$ matrices A and B,

$$\mathrm{tr}(AB) = \mathrm{tr}(BA).$$

Student Number:									

(b) Prove or disprove: There are $n \times n$ matrices A and B such that $AB - BA = I_n$.

(c) Let A be any skew-symmetric matrix. Find $\mathrm{tr}(A)$.